

Total No. of Questions :6]

SEAT No. :

P10

TE/Insem./APR-13

[Total No. of Pages : 2

T.E (Civil)

301009 : FOUNDATION ENGINEERING

(2015 Pattern) (Semester II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Answer to the two sections should be written in separate books.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables, slide rule, Molli's charts, electronics pocket calculator and steam tables is allowed
- 5) Assume suitable data if necessary
- 6) Neat diagrams must be drawn wherever necessary.

- Q1) a)** Explain in detail Seismic Refraction Method of soil investigation. [5]
- b) Write a note on purpose and planning of subsurface exploration. [5]

OR

- Q2) a)** Explain with neat sketch SPT and explain how this test is performed on the field. [4]
- b) A sampling tube of 72 mm diameter and 5 mm thick. It is fitted with cutting edge. The inside diameter of cutting edge is flushed with sampling tube. The cutting edge is 6 mm thick. Compute inside clearance, outside clearance, and area ratio. Comment on sample collected by tube. [6]

- Q3) a)** Explain in detail various modes of bearing capacity failures along with suitable diagram. [5]
- b) A square foundation is 1.5m x 1.5m in plan. The soil supporting the foundation has a friction angle $\Phi=20^\circ$, $C=15.2 \text{ kN/m}^2$. The unit weight of soil, γ , is 17.8 kN/m^3 . Determine the allowable gross load on the foundation with a factor of safety of 4. Assume that depth of foundation is 1m and that general shear failure occurs in soil when $N_c=17.69$, $N_q=7.44$, $N_\gamma=3.64$ [5]

OR

P.T.O.

Q4) a) Explain in detailed with neat sketch bearing capacity evaluation by plate load test. [5]

b) Write a note on effect of water table on bearing capacity. [5]

Q5) a) Enlist different causes of settlement and explain any one in detail. [5]

b) In a consolidation test the void ratio decreases from 0.7 to 0.65 when the load was changed from 50 kN/m² to 100 kN/m². Compute compression index and coefficient of volume change. [5]

OR

Q6) a) Explain with neat sketch laboratory consolidation test. [5]

b) A soil sample having void ratio 0.6 decreases by 0.1 when subjected to increasing load of 200 kN/m² to 275 kN/m². Calculate the coefficient of compression and coefficient of volume compressibility. [5]

